

Kiloparsec-scale AGN Outflows and Feedback in Merger Free Galaxies

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2021.10.28

Recent observations and simulations have challenged the long-held paradigm that mergers are the dominant mechanism driving the growth of both galaxies and supermassive black holes.

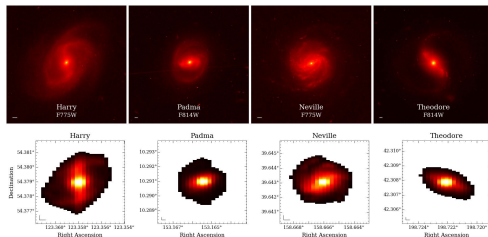
Old Galaxy growth mainly occurs through mergers

New Galaxy growth mainly occurs through non-merger “secular processes” rather than by mergers

- Millenium simulations: the majority of bulge mass is built through disk instabilities, Parry et al. (2009)
- Observations: only 27% of star formation is triggered by mergers at $z \sim 2$, Kaviraj et al. (2013)

Observations

A significant bulge component still forms even in a minor merger
Select 4 disk-dominated galaxies with luminous, unobscured Type 1 AGN at $z < 0.1$

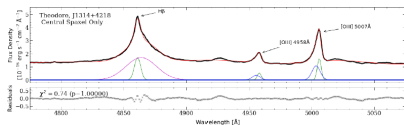
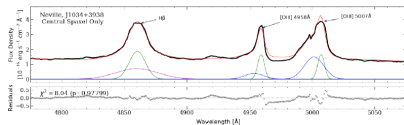
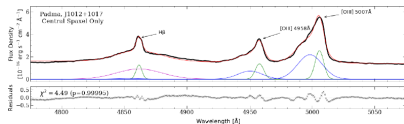
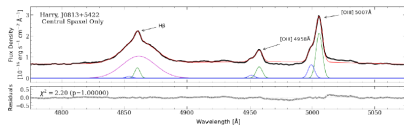


All four galaxies have little or no bulge component.
Harry and Theodore show a strong bar feature, Neville a weak bar feature, and Padma a bar/enslaved feature.

Observations

- narrow $H\beta$ shows the galaxy structure
- narrow O_{III} ionized by the central AGN
- broad O_{III} ionized by the AGN outflow

Observations



Flux Density - Wavelength

black Spectrum

red Fittings

cross Residuals

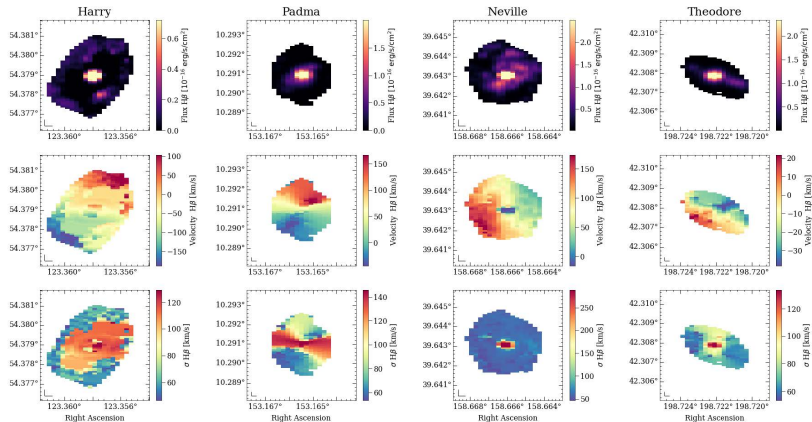
O_{III}

- narrow
- broad

H β

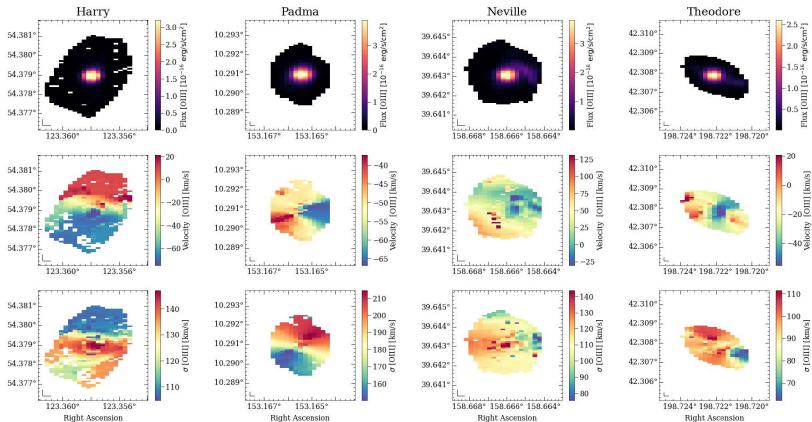
- narrow
- broad
- broad line region

Observations



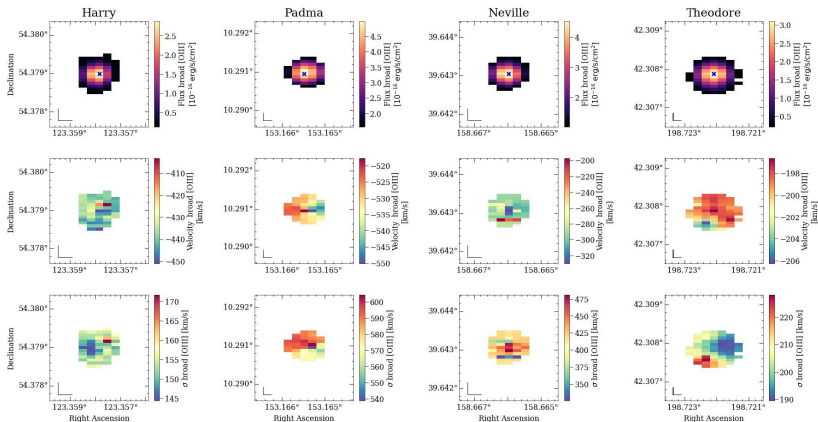
narrow H β integrated flux, velocity, velocity dispersion

Observations



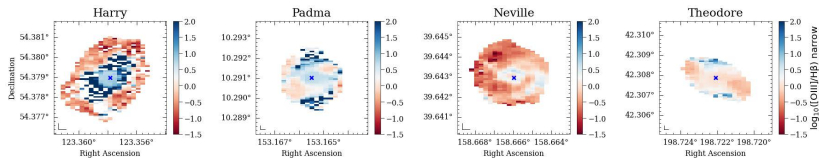
narrow O_{III} integrated flux, velocity, velocity dispersion

Observations

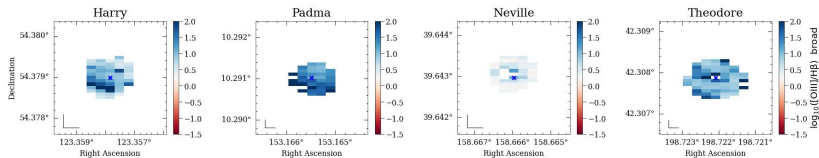


brod O_{III} integrated flux, velocity, velocity dispersion

Observations



narrow $O_{III}/H\beta$ emission



broad $O_{III}/H\beta$ emission

Calculations

- Outflow luminosity from broad O_{III} gives gas mass in the outflow:

$$M_{\text{O}_{\text{III}}} = 8 \times 10^7 M_{\odot} \left(\frac{\overline{n_e^2}/\overline{n_e^2}}{10^{[\text{O}/\text{H}] - [\text{O}/\text{H}]_{\odot}}} \right) \left(\frac{L[\text{O}_{\text{III}}]}{10^{44} \text{ erg/s}} \right) \left(\frac{n_e}{500 \text{ cm}^{-3}} \right)^{-1}$$

assuming that $n_e = 500 \text{ cm}^{-3}$ and $[\text{O}/\text{H}] = [\text{O}/\text{H}]_{\odot}$

- Maximum velocity of the outflow:

$$v_{\text{O}_{\text{III}}} = |\Delta v_{\text{max}}| + 2\sigma_{\text{broad,max}}$$

Calculations

- Physical extent of the outflow r_{max} (deconvolved by seeing)
- Timescale of the outflow $t_{out} = \frac{r_{max}}{v_{OIII}}$
- Mass, kinetic energy and momentum outflow rate:

$$\dot{M} = B \frac{M_{OIII}}{t_{out}}, \quad \dot{E} = \frac{1}{2} \dot{M} v_{OIII}^2, \quad \dot{P} = \dot{M} v_{OIII}$$

assuming geometry factor $B = 1$

- Black hole accretion rates $\dot{m} = \eta L_{bol} c^2$ where the radiative efficiency $\eta = 0.15$ and L_{bol} using the W3 12 μm band

Results

Name	$\log_{10}$ [M _{BH} /M _☉]*	$\log_{10}$ [L _{OIII} /erg s ⁻¹]	λ_{Edd}^*	$\dot{m}^*$ [M _☉ yr ⁻¹]	$\log_{10}$ [M _{OIII} /M _☉]	$v_{\text{max}, [\text{OIII}]}$ [km s ⁻¹]	r_{max} [kpc]	$\dot{M}_{\text{out}}$ [M _☉ yr ⁻¹] [†]	t_{out} [Myr]
Harry	$6.56^{+0.13}_{-0.12}$	41.2 ± 1.2	$0.08^{+0.33}_{-0.02}$	$0.02^{+0.04}_{-0.01}$	5.1 ± 0.1	836 ± 28	0.6 ± 0.3	0.19 ± 0.09	0.6 ± 0.3
Padma	$7.62^{+0.14}_{-0.14}$	42.2 ± 0.2	$0.20^{+0.45}_{-0.09}$	$0.07^{+0.4}_{-0.3}$	6.03 ± 0.09	1710 ± 6	2.4 ± 0.4	0.7 ± 0.1	1.4 ± 0.2
Neville	$6.30^{+0.12}_{-0.12}$	41.6 ± 0.4	$0.86^{+2.90}_{-0.26}$	$0.07^{+0.11}_{-0.04}$	5.5 ± 0.1	1316 ± 29	2.1 ± 0.3	0.18 ± 0.03	1.6 ± 0.2
Theodore	$6.73^{+0.11}_{-0.11}$	41.6 ± 0.6	$0.77^{+1.68}_{-0.35}$	$0.06^{+0.04}_{-0.02}$	5.4 ± 0.2	675 ± 18	1.3 ± 0.4	0.12 ± 0.04	1.9 ± 0.6

- $\overline{M}_{\text{OIII}} = 5.5 \pm 0.2 M_{\odot}$
- $\overline{\dot{M}} = 0.3 \pm 0.1 M_{\odot}/\text{yr}$
- $\overline{r_{\text{max}}} = 1.6 \pm 0.4 \text{ kpc}$, $\sim 25\%$ of the galaxy Petrosian radius

Discussion

- Bars and spiral arms would be capable of driving inflows which could sustain both the SMBH growth ($0.02 \sim 0.07 M_{\odot}/\text{yr}$) and an outflow of AGN ($0.2 \sim 0.7 M_{\odot}/\text{yr}$) for an extended period of $0.6 \sim 1.9 \text{ Myr}$
- Secular processes are responsible for the majority of SBMH growth, whereas mergers are responsible for the majority of outflowing material and the subsequent feedback on the galaxy
- The average ratio $\log_{10}(c\dot{P}/L_{\text{bol}}) = -0.91 \pm 0.08$ suggests that these outflows are momentum-conserving
- The average v_{OIII} of the 4 targets is ~ 30.5 times larger than escape velocity, which will cause AGN feedback

Discussion

- The outflow rates in Bae et al. (2017) are ~ 51 times larger than in this article, but SMBH accretion rates are ~ 3 times lower, this may cause by a spin up of SMBHs due to a secular feeding mechanism
- A higher spatial resolution IFU study and a larger sample of disk-dominated galaxies would allow for a more detailed study

Conclusion

- Secularly powered outflows are typical of low-redshift Type 1 AGNs and they have momentum conserving outflows
- Secular processes are responsible for the majority of SMBH growth over cosmic time

Thank you for listening!